Open
Computing
Facility

Lecture 6

lemurseven & storce

Web Servers

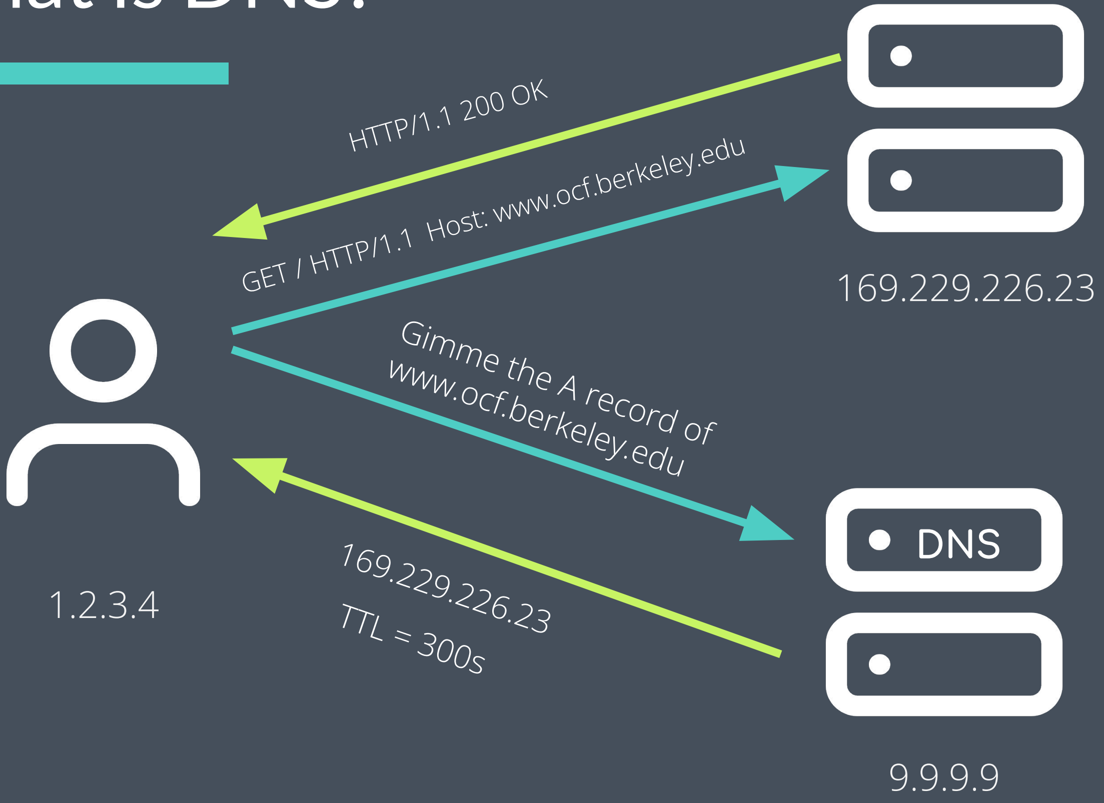
What happens when we visit a website?

Overview

- Domain Name System
- Web servers
- Load balancing

Domain Name System

What is DNS?



HTTP/1.1 200 OK

GET / HTTP/1.1 Host: www.ocf.berkeley.edu

169.229.226.23

Gimme the A record of www.ocf.berkeley.edu

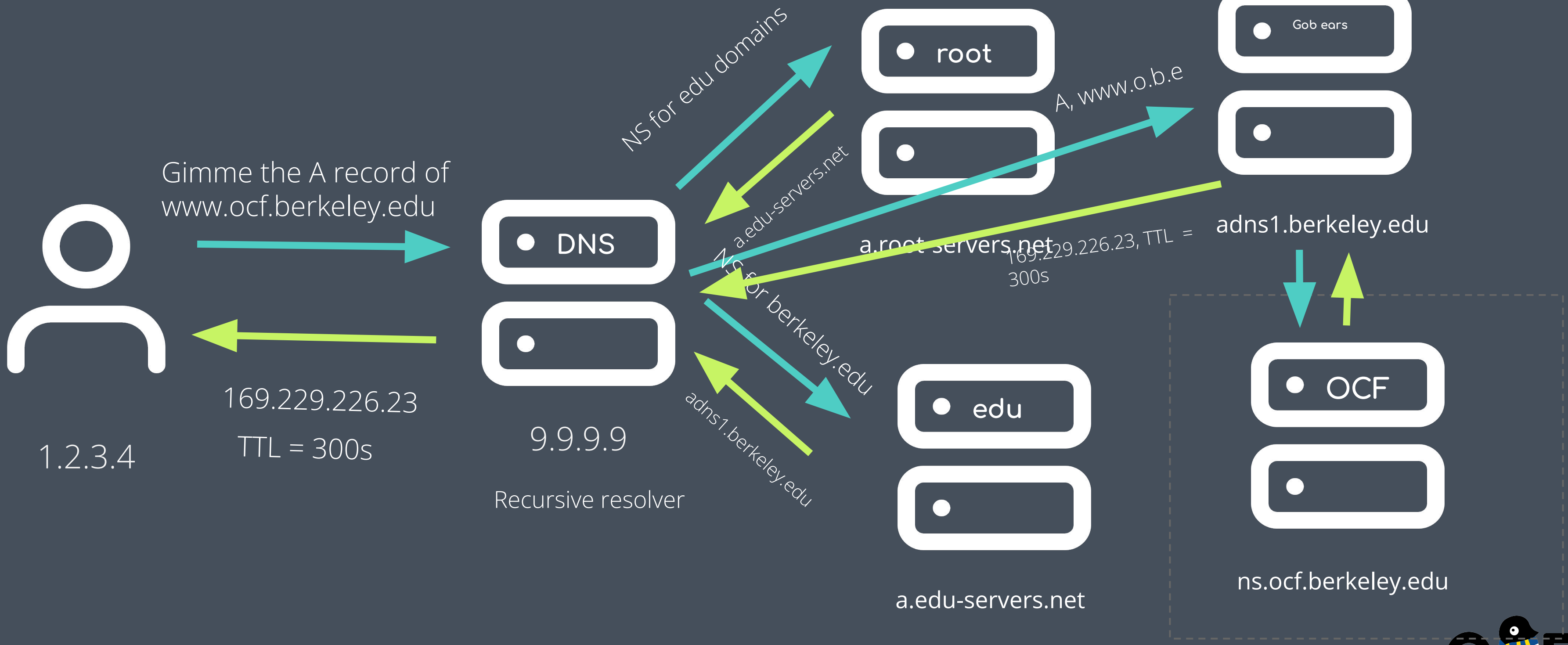
DNS

1.2.3.4

169.229.226.23
TTL = 300s

9.9.9.9

What is actually happening?



Gimme the A record of
www.ocf.berkeley.edu

1.2.3.4

169.229.226.23

TTL = 300s

DNS

9.9.9.9

Recursive resolver

NS for edu domains

a.edu-servers.net

root

a.root-servers.net

NS for berkeley.edu

adns1.berkeley.edu

edu

a.edu-servers.net

A, www.o.b.e

169.229.226.23, TTL =
300s

Gob ears

adns1.berkeley.edu

OCF

ns.ocf.berkeley.edu

Types of DNS records

- A: returns an IPv4 address (e.g. 74.125.142.147)
- AAAA: returns an IPv6 address (e.g. 2607:f140:0:32::70)
- CNAME: returns the canonical domain name (e.g. uptime.ocf.io points to stats.uptimerobot.com)
- MX: redirects email to a mail server (e.g. MX ocf.b.e points to aspmx.l.google.com etc.)
- NS: stores the authoritative name server for a domain (e.g. ocf.io's NS record points to ns1.o.b.e)

More types of DNS records

- TXT: contains information about the domain (e.g. site verification, etc.)
- SRV: specifies a host and port for specific services
- SOA: stores administrative information about a domain (such as the email address of the admin, when the domain was last updated, and how long the server should wait between refreshes)

The DNS records of OCF

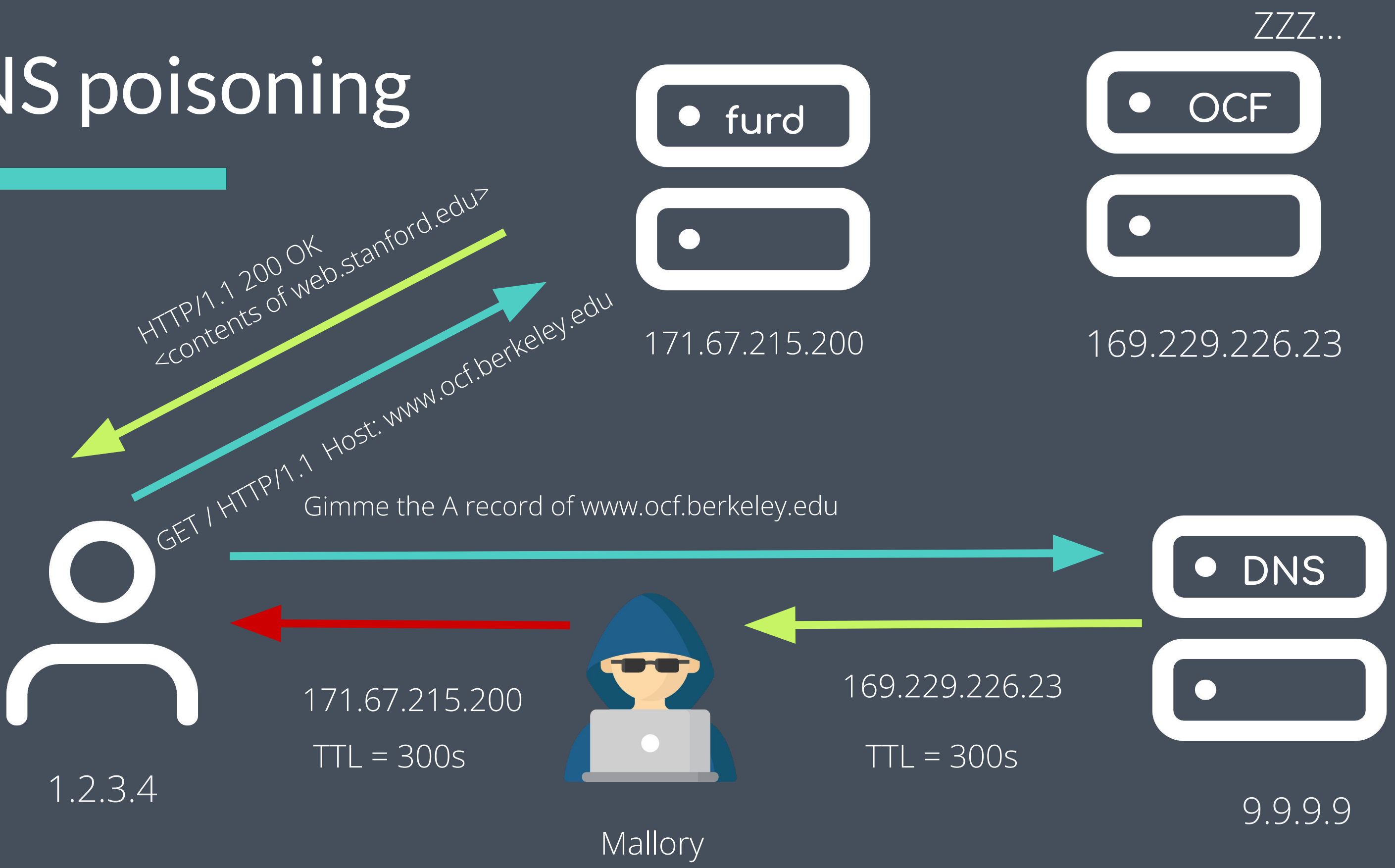
https://github.com/ocf/dns

```
death             IN A        169.229.226.23
death             IN AAAA     2607:f140:8801::1:23
www               IN CNAME    death
fallingrocksIN A        169.229.200.70
fallingrocksIN AAAA     2607:f140:0:32::70
apt               IN CNAME    fallingrocks
mirrors           IN CNAME    fallingrocks
lb                IN A        169.229.226.79
lb                IN AAAA     2607:f140:8801::1:79
auth              IN CNAME    lb
...
```

TTL record

- Time to live
- Tells a DNS server or the local resolver how long it should keep the record in its cache
- Longer TTLs can speed up DNS resolution but causes updates to the zone to take longer to propagate to users

DNS poisoning



ZZZ...

furd

OCF

HTTP/1.1 200 OK
<contents of web.stanford.edu>

GET / HTTP/1.1 Host: www.ocf.berkeley.edu

171.67.215.200

169.229.226.23

Gimme the A record of www.ocf.berkeley.edu

DNS

171.67.215.200
TTL = 300s

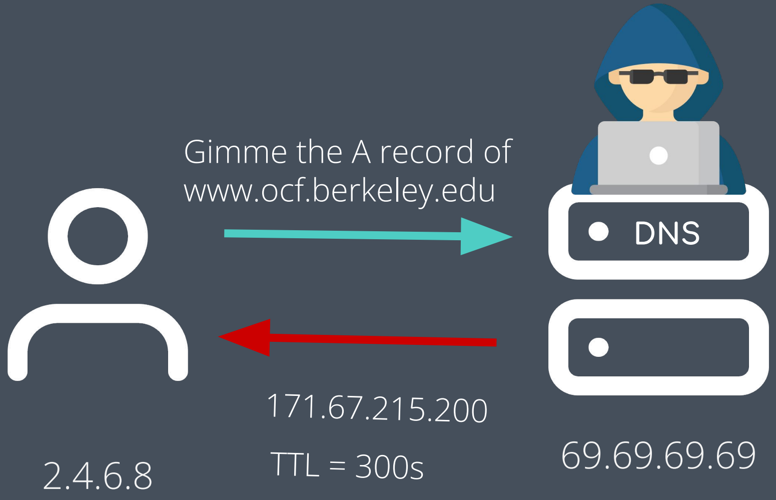
169.229.226.23
TTL = 300s

1.2.3.4

Mallory

9.9.9.9

More DNS poisoning



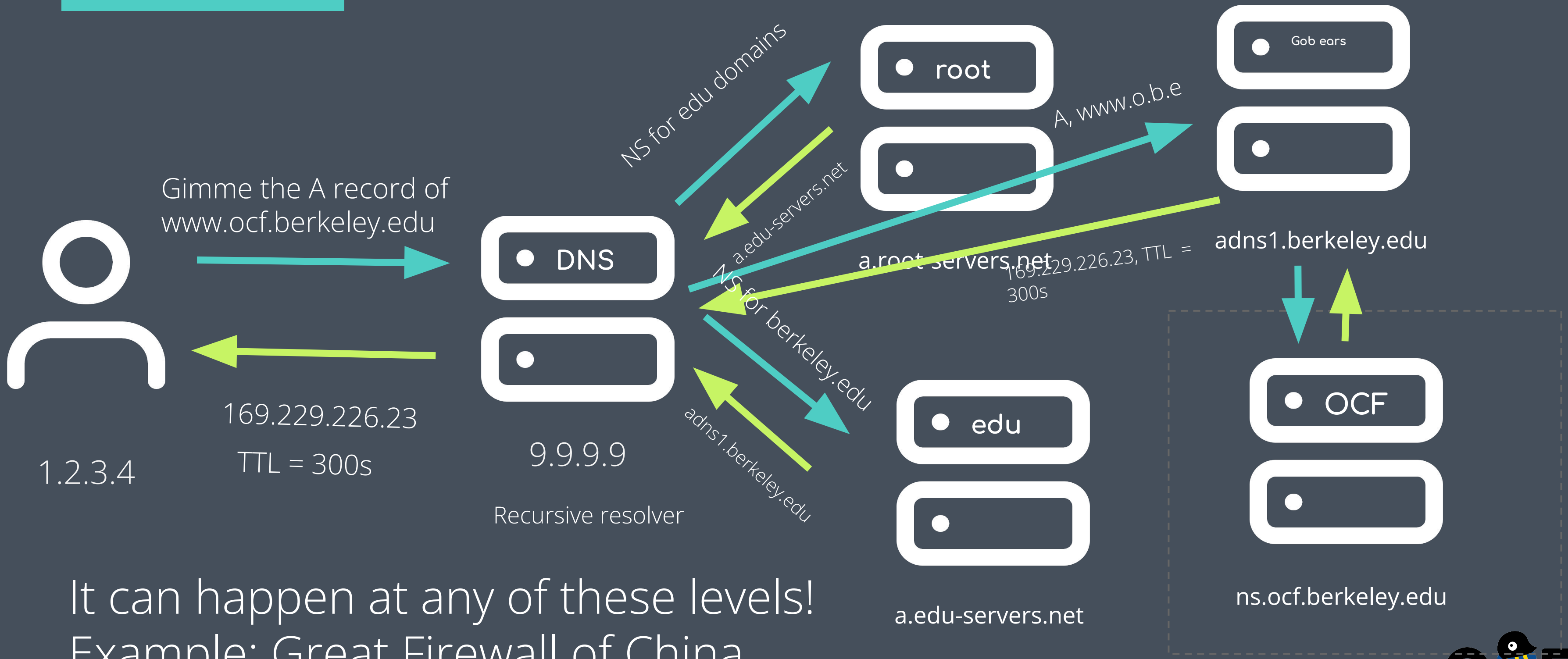
Gimme the A record of www.ocf.berkeley.edu

171.67.215.200

TTL = 300s

DNS
2.4.6.8
69.69.69.69

More DNS poisoning



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Gob ears

adns1.berkeley.edu

NS for berkeley.edu

adns1.berkeley.edu

edu

a.edu-servers.net

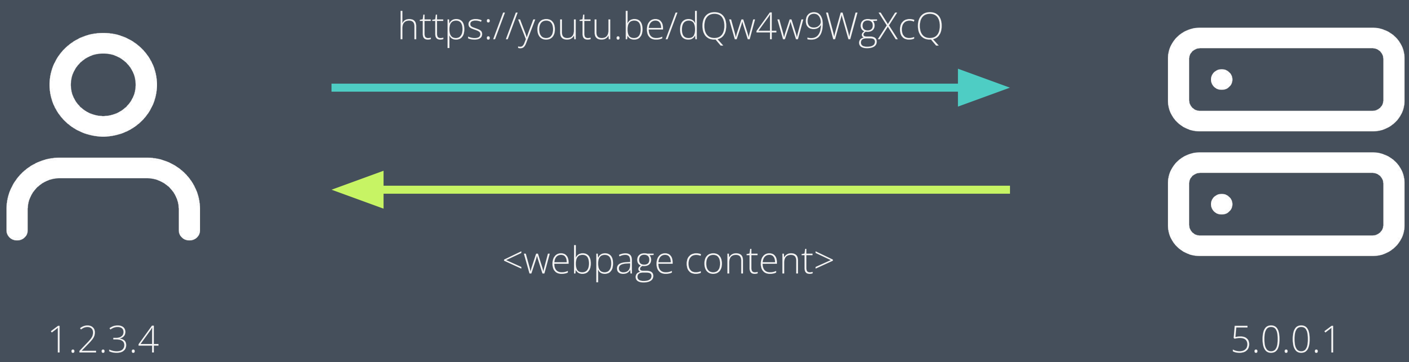
OCF

ns.ocf.berkeley.edu

It can happen at any of these levels!
Example: Great Firewall of China

Web Servers

What do web servers do?



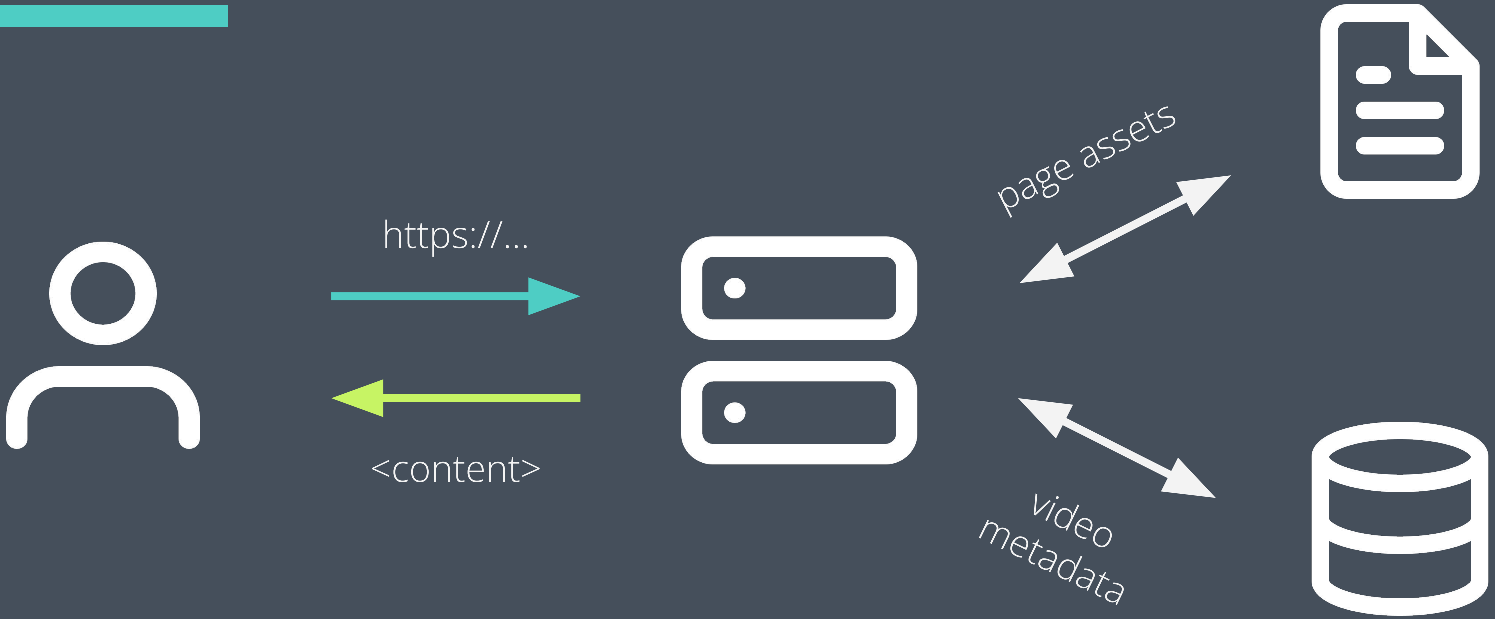
https://youtu.be/dQw4w9WgXcQ

<webpage content>

1.2.3.4

5.0.0.1

What do web servers do?



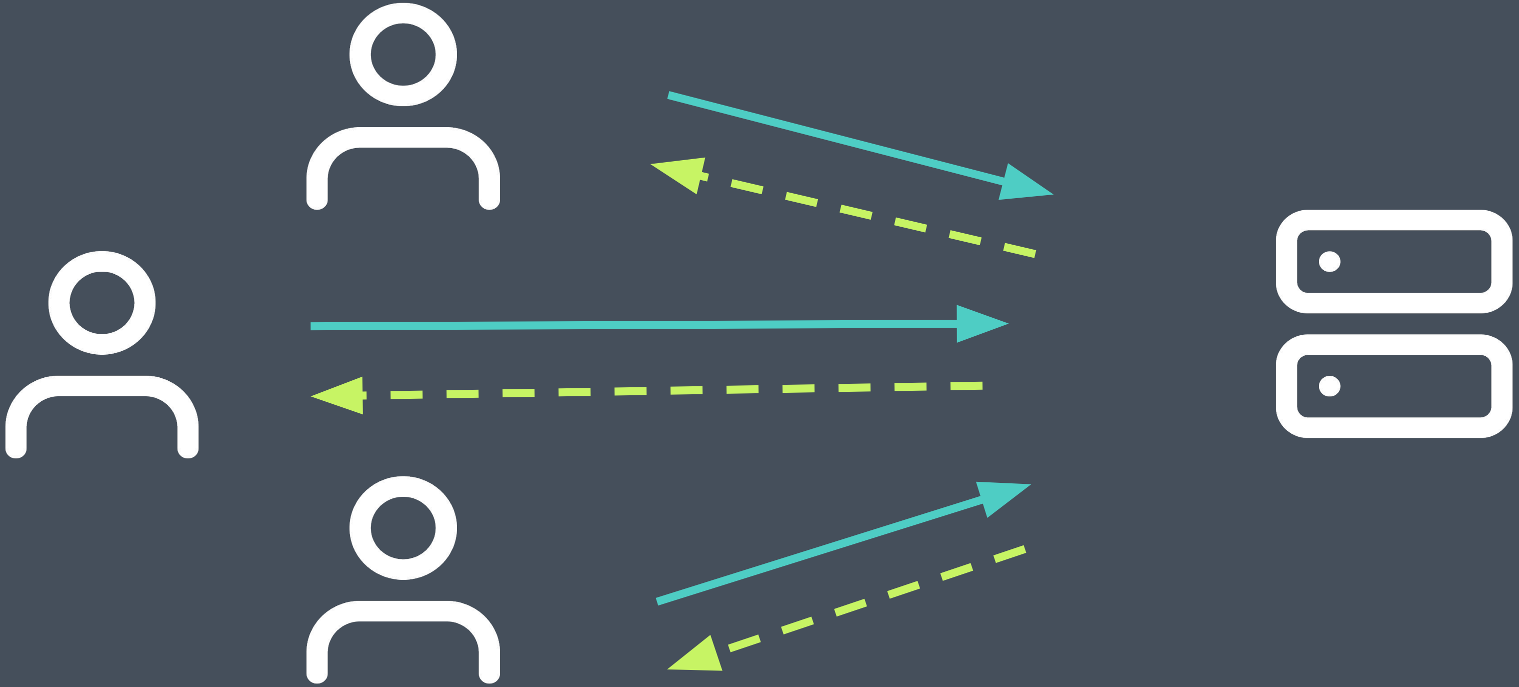
https://...

<content>

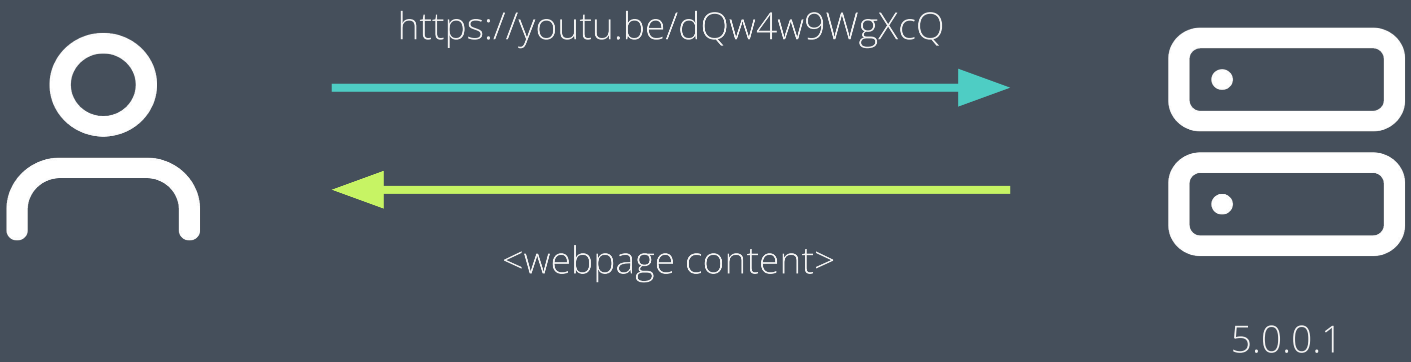
page assets

video metadata

How do web servers scale?



How do web servers scale?



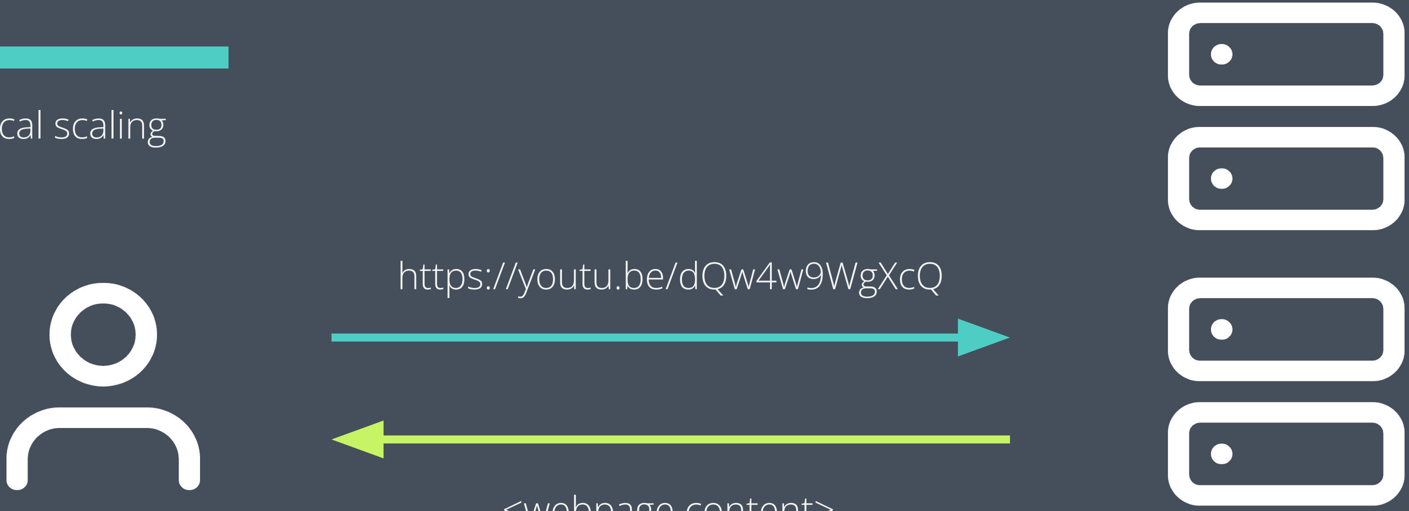
https://youtu.be/dQw4w9WgXcQ

<webpage content>

5.0.0.1

How do web servers scale?

vertical scaling



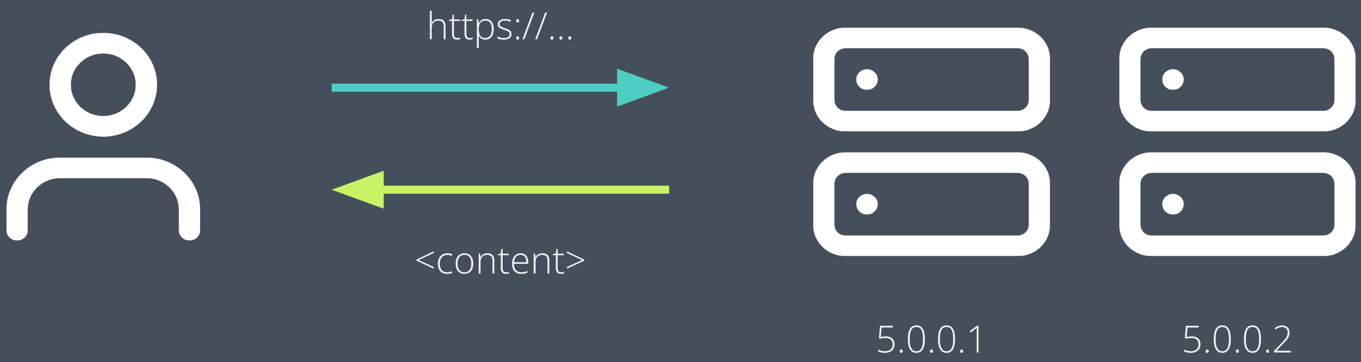
https://youtu.be/dQw4w9WgXcQ

<webpage content>

5.0.0.1

How do web servers scale?

horizontal scaling



https://...

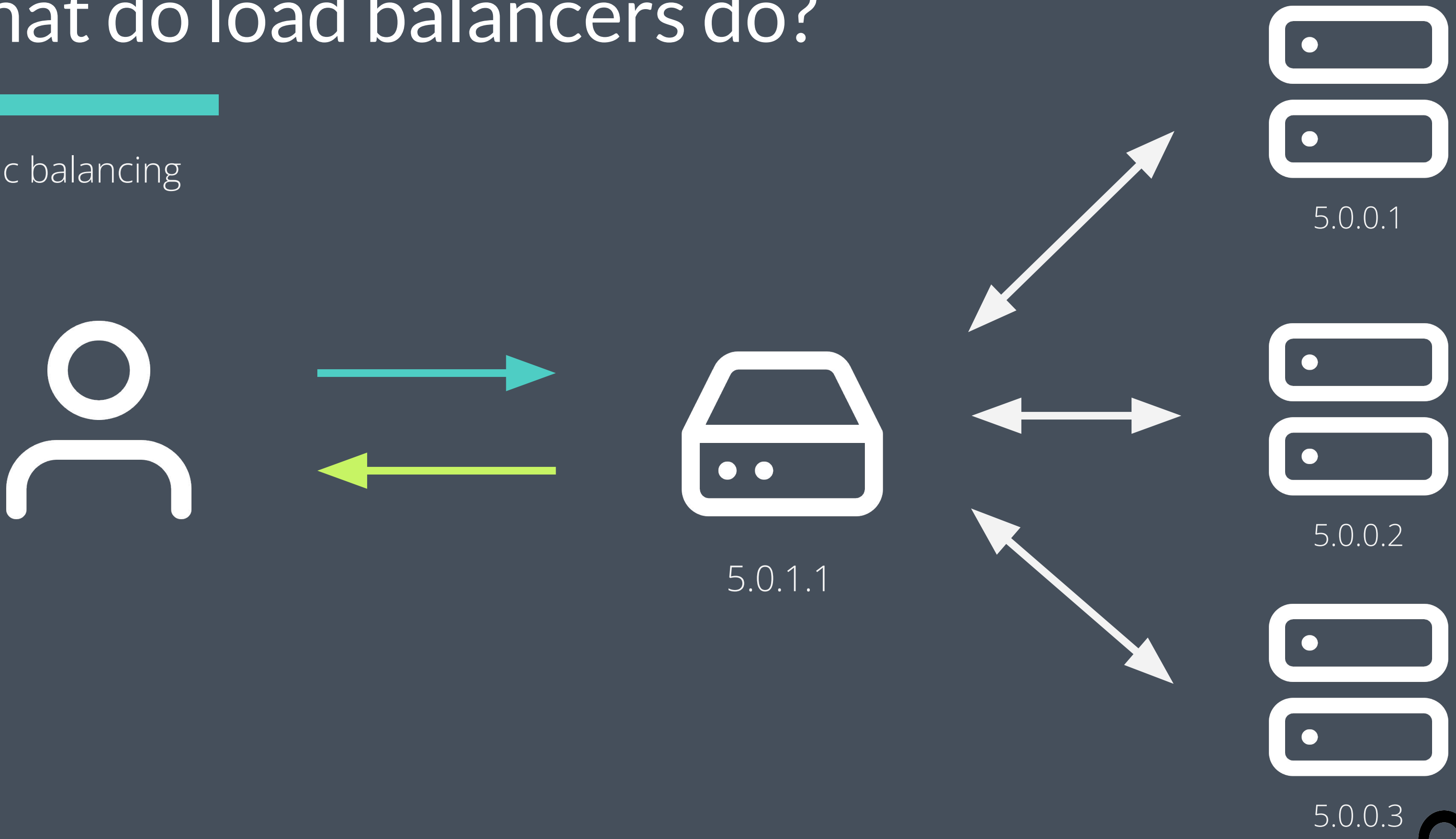
<content>

5.0.0.1

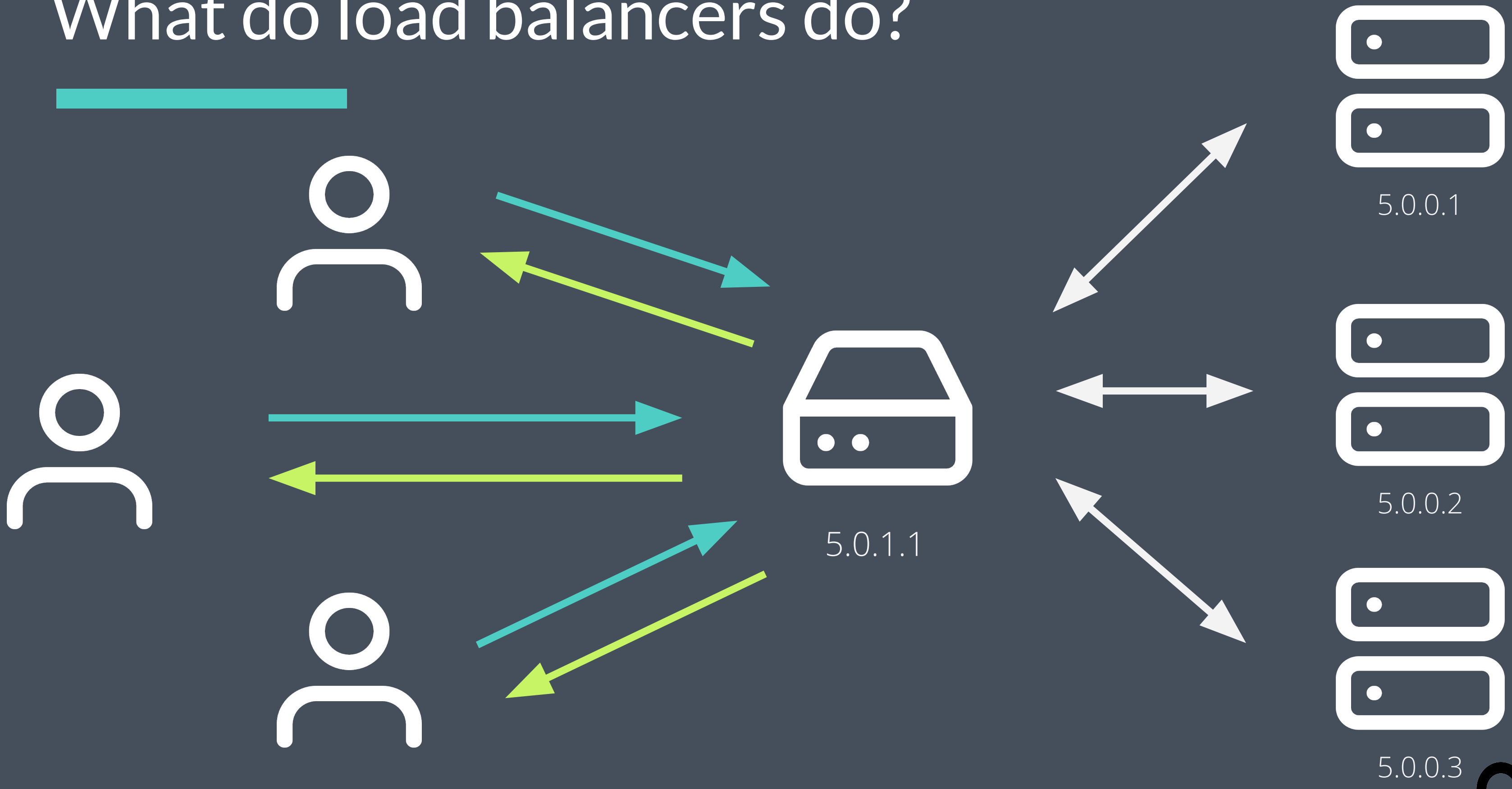
5.0.0.2

What do load balancers do?

traffic balancing

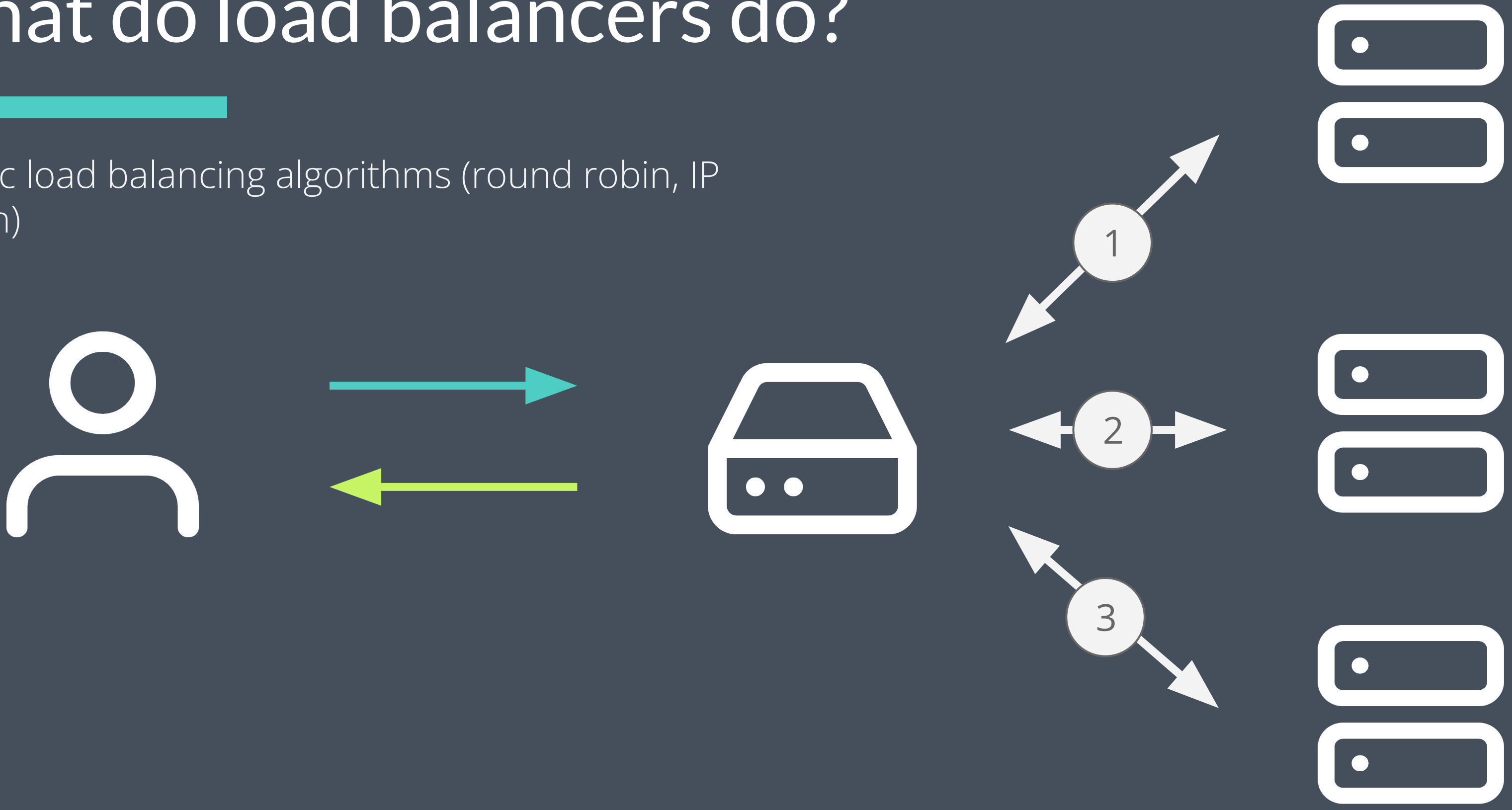

5.0.0.1
5.0.0.2
5.0.1.1
5.0.0.3

What do load balancers do?


5.0.1.1
5.0.0.1
5.0.0.2
5.0.0.3

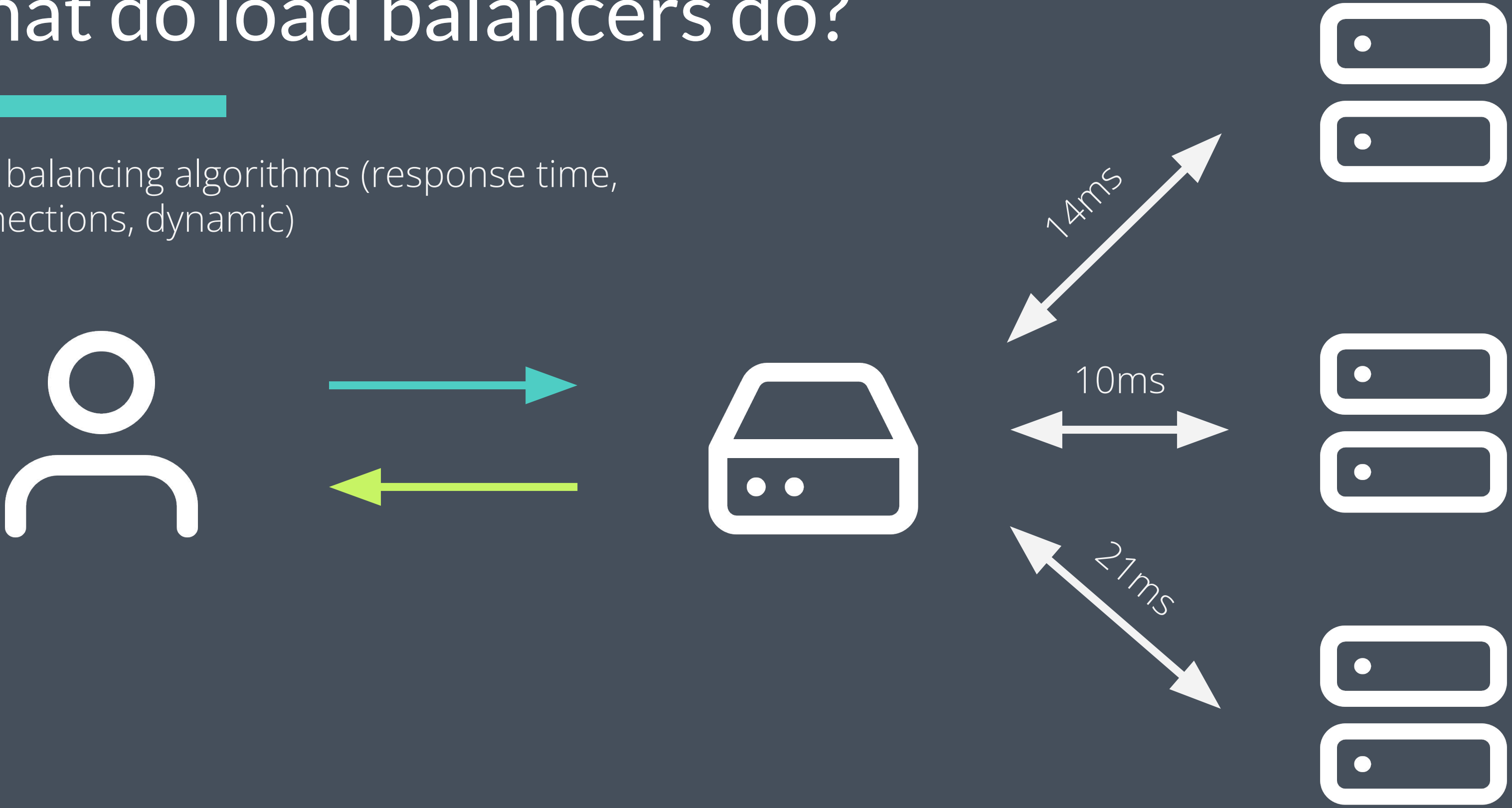
What do load balancers do?

Static load balancing algorithms (round robin, IP Hash)



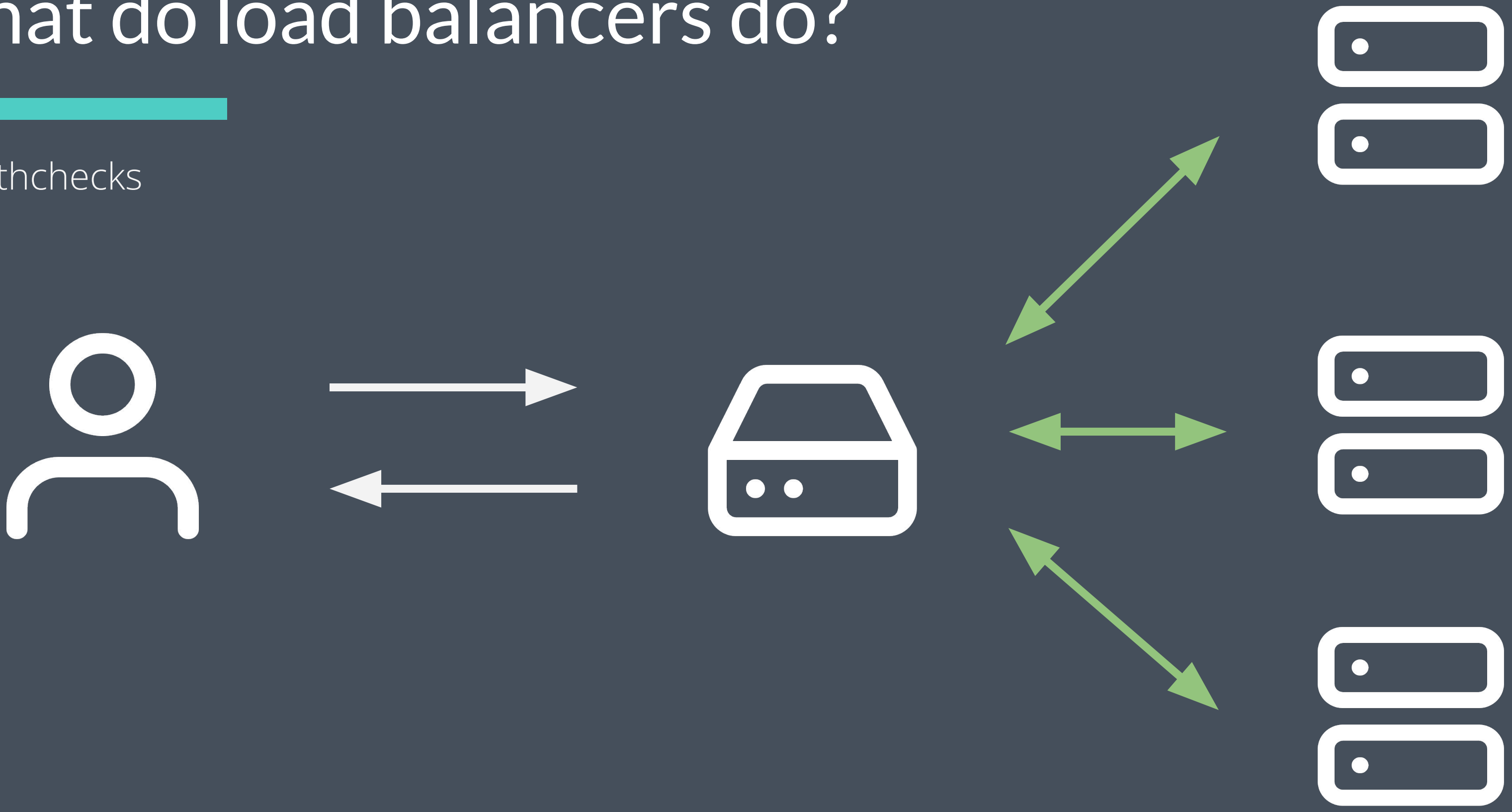
What do load balancers do?

load balancing algorithms (response time, connections, dynamic)

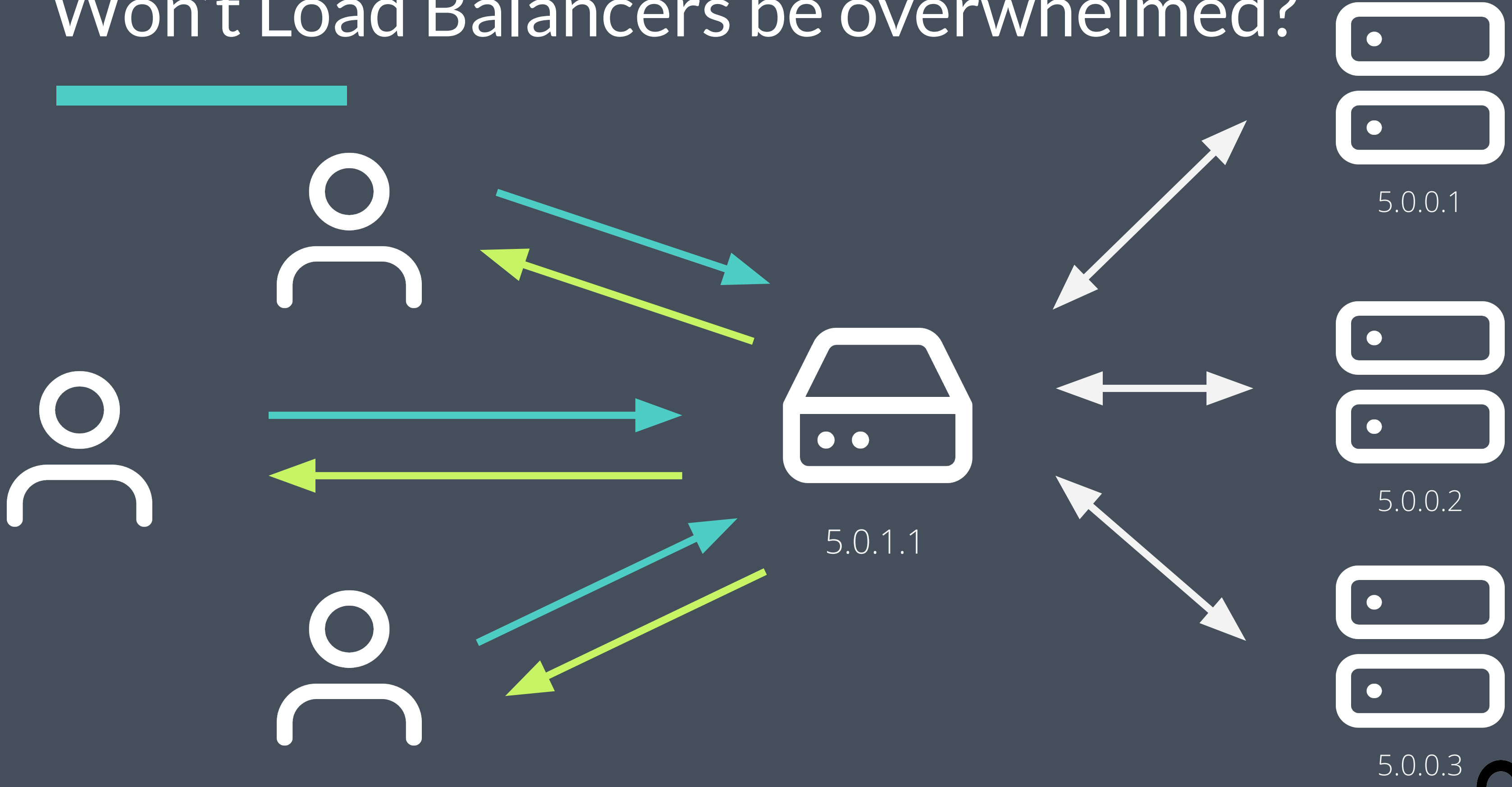

14ms
10ms
21ms

What do load balancers do?

healthchecks



Won’t Load Balancers be overwhelmed?


5.0.0.1
5.0.0.2
5.0.1.1
5.0.0.3